Cohesion-Gated Taxonomy Construction: Label-Free Geometric Confidence for Allocating LLM Inference, Adversarially Audited

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Abstract

Organizing an unlabeled document corpus costs $O(N)$ decisions, whether a human or a language model makes them. We study a pipeline that spends local computation exhaustively and frontier-model inference selectively: embed every document locally, cluster, let an LLM name only the clusters, and use *whole-cluster cohesion* (mean cosine of members to their centroid, computed with zero model calls) to decide which clusters receive one propagated cluster-level extraction and which receive exhaustive per-item reads. We state the method's claims at the right altitude and attack them. The *signal* claim survives every test we ran: cohesion predicts cluster purity at Spearman +0.74, replicates across three corpora and five seeds, survives a cluster-size control (partial correlations +0.48 to +0.71), and the cohesion gate beats a random gate at equal budget on tasks never used to tune it. The *savings* claim does not survive in unconditional form. On LEDGAR governing-law extraction the gated hybrid reaches parity with exhaustive reading at roughly 2–5× fewer calls; a pre-registered falsification test on two untouched task–corpus pairs shows this ratio is a *corpus property, not a method constant*: parity is borderline at 2× on 20 Newsgroups and unattained at any budget below 100% on Amazon categories. Across the three tasks the attainable parity budget tracks the gap between the clustering's purity ceiling and per-item accuracy (4.8, 9.8, and 20.7 points respectively), making the popular "k× fewer calls" framing meaningless without stating the corpus's redundancy. An end-to-end audit in which the system must first *find* the concept with its own taxonomy shows the isolated-benchmark accuracy overstates production accuracy: routing recall 0.88 and precision 0.75 reduce correct-value coverage to 86.4%, and 89% of intruder items receive a confidently propagated wrong value, motivating a deterministic validator on all propagated values. We additionally report a hyperbolic-geometry arc with a methodological lesson: every negative hyperbolic result we produced traced to gradient optimization that fails a known-answer control, while a deterministic Sarkar construction embeds an 82,115-node hierarchy in *two* dimensions with higher tree-distance fidelity ($\rho = +0.864$) than our gradient-trained Euclidean baseline in eight (+0.610); and an item-level audit that falsifies our own hyperbolic anomaly score against a plain cosine baseline, yielding a better two-queue replacement. All claims carry pre-registered falsifiers; negative results and retractions are reported at equal weight.

1 Introduction

Turning a pile of documents into structure is bottlenecked by per-item attention. Reading every item with a frontier language model is accurate but linear in cost; human labeling is linear in scarcer currency. Most of that linear effort is spent on repetitive, internally consistent material.

We evaluate a two-stage architecture on this problem. An exhaustive shallow stage runs locally and at negligible cost: embed every document with a small local model, cluster, and compute geometric evidence about every cluster. A selective deep stage spends expensive inference (LLM calls, human review) only where the geometry indicates uncertainty. The routing signal is *cohesion*: the mean cosine similarity of a cluster's members to the cluster's unit centroid. Cohesion costs no model calls, so the budget decision itself is free.

The embed–cluster–name pipeline is well-trodden prior art (Section 2). The specific mechanism we evaluate, and did not find in the literature, is using whole-cluster cohesion, computed *before any item-level LLM call*, to decide which clusters get one cluster-level extraction propagated to all members and which get exhaustive per-item inference. We phrase this as "not found in the literature we reviewed," not as a novelty claim.

This paper has a second, methodological subject: what it takes to believe your own numbers. The initial results (Section 5) were promising, and several were wrong in ways invisible from inside the experiment. We therefore treat the audit of our own claims as first-class content (Sections 5.2–5.9): an oracle-contamination test, an out-of-sample calibration test, a ground-truth audit, a seed-stability test, and an end-to-end test in which the system must find the concept before extracting from it. Two audits weakened headline claims; two strengthened them; one exposed a production failure mode the benchmark structurally could not see. Section 6 reports our negative results and the three-stage correction of our hyperbolic-geometry claims, which reduces to a transferable rule: *never trust a hyperbolic embedding that has not passed a known-answer control; training loss is not a health signal.*

We evaluate four claims, each with a falsifier declared before the deciding experiment ran:

B1 *Signal.* Cohesion, computed with zero model calls, ranks clusters by propagation safety, and the cohesion gate beats a random gate at equal budget on tasks never used for development. *Falsifier:* no accuracy separation by cohesion, or gate $\leq$ random, on a held-out task.

B2 *Frontier.* Gated routing reaches parity with read-everything (paired CI lower bound ≥ -2.0 points) at some budget materially below 100%, with the parity budget task-dependent. *Falsifier:* no sub-100% budget reaches parity on a held-out task.

B3 *Boundary.* The mechanism requires the target attribute to dominate the text; on incidental attributes it loses to a majority baseline. (Deliberately sought, and confirmed.)

B4 *Leverage.* Naming k clusters organizes $N \gg k$ documents at useful purity across domains.

The outcome: B1, B3, and B4 survive; B2 is falsified in unconditional form and survives only as a conditional claim whose condition we measure (Section 5.4).

Contributions:

1. A measured cost/quality frontier for cohesion-gated extraction on LEDGAR, with real-call, out-of-sample, and multi-seed audits (parity from a 40% read budget with everything real; gate beats random at every budget), and a pre-registered two-task falsification showing

the attainable savings ratio is a corpus property governed by the purity-ceiling gap, not a method constant.

2. A three-corpus, five-seed validation that cohesion predicts cluster purity (ρ from +0.52 to +0.74) and is not a cluster-size artifact (partial ρ from +0.48 to +0.71), plus evidence that the multivariate alternative overfits (leave-one-out R^2 collapse).

3. An end-to-end audit separating *finding* from *extracting*, quantifying how benchmark isolation subsidizes accuracy (96% → 86.4% coverage) and exposing an 88.9% false-assertion rate on intruders, with a zero-cost deterministic mitigation.

4. An item-level falsification of our own hyperbolic anomaly score against a cosine baseline (AUC 0.763 vs. 0.814), an ablation showing its radius term is both cosmetic and sign-inverted relative to where novel concepts live, and a two-queue Euclidean replacement that raises novel-concept precision@40 from 5% to 55%.

5. A replication lesson for hyperbolic embeddings: our gradient-trained Poincaré arm failed the WordNet positive control that the literature passes, invalidating six of our own negative comparisons, while Sarkar's optimization-free construction reproduced the published dimension-efficiency result (2-D beating our 8-D Euclidean).

Code, per-experiment scripts with declared falsifiers in their docstrings, and machine-readable results accompany the paper.[1]

2 Related work

Embed, cluster, name. BERTopic [1], TopicGPT [2], TnT-LLM [3], and Clio [4] establish the pattern of clustering embeddings and using an LLM to label clusters; commercial systems (Nomic Atlas, Relativity Analytics) have shipped variants for years. Querying an LLM on cluster representatives and propagating to members appears in [5] and, in the label-acquisition setting, in [6].

Confidence-gated cascades. FrugalGPT [7] routes queries between models by predicted difficulty, but its gate requires at least one model call per item; our gates (cohesion, item-margin) cost zero calls and therefore operate below the cascade's cost floor.

Internal cluster validity. Cohesion is a member of the silhouette family [8]; published internal-vs-external validity correlations of $r \approx 0.65$–0.66 bracket our +0.74, so the existence of the correlation is not novel; its use as a pre-hoc inference-budget allocator is the part we did not find. Geometry-driven budget allocation appears in active learning as TypiClust [9] and ProbCover [10].

Constrained clustering. Human anchors as structural attractors follow seeded/constrained clustering [11] and anchored topic models.

Hyperbolic embeddings. Poincaré embeddings [12] and the combinatorial constructions of Sarkar [14] as analyzed by Sala et al. [13] motivate our hierarchy-representation experiments; Sala et al. explicitly document the unreliability of gradient optimization in the Poincaré ball, which our Section 6.2 reproduces exactly.

Anisotropy. Our mean-centering result is step one of All-but-the-Top [15].

[1] `https://github.com/cptntrps/cohesion-gated-taxonomy`

3 Method

3.1 Pipeline

Documents are embedded locally (`nomic-embed-text`, 768-d, unit-normalized; embeddings are cached, so the corpus never leaves the host and marginal cost is zero). A divisive KMeans tree is built top-down. An LLM names only clusters, from the k=10 most central members plus the ancestor chain; naming failures are surfaced, never backfilled. Gold labels are used exclusively for post-hoc scoring, never in clustering, naming, routing, or stopping.

3.2 Geometric evidence

For a cluster with member matrix V_m (unit rows) and unit centroid c:

$$\text{coh}(V_m) = \frac{1}{|m|} \sum_i \langle v_i, c \rangle.$$

Multimodality is the cosine silhouette of the best 2-means split of the cluster: high values indicate a real seam (two concepts merged); low values indicate coherence, however coarse.

3.3 Two modes with different loss functions

MAP answers "where does each item belong in a taxonomy I have?" It splits to a fixed depth to maximize assignment purity; its failure mode is wrong assignment. EMERGE answers "what concepts exist, given no taxonomy?" It splits *only* where multimodality exceeds a seam threshold and stops when unimodal; its failure mode is the *false merge* of distinct concepts, and a coarse-but-coherent node is correct output, not error. Scoring EMERGE against a reference taxonomy is a category error: on 20 Newsgroups the system merged `rec.autos` with `rec.motorcycles`, arguably a better taxonomy than the reference, scored as error by purity.

3.4 Human anchors

A human rename or new concept name is embedded into the document space and becomes an attractor: every document joins its nearest anchor if cosine ≥ 0.60, the remainder falls back to discovery clustering, and anchored clusters keep the human name. An anchor no document matches remains visible and empty. Corrections are therefore structural, not cosmetic, and the interactive loop (observe $\rightarrow$ name $\rightarrow$ re-cluster) lets a domain expert encode a folk taxonomy without taxonomy expertise.

3.5 Cohesion-gated extraction

Given a frozen clustering and an extraction field, the system makes one real LLM call per cluster (on six central representatives) and propagates the answer to members. Clusters in the lowest cohesion quantile (the *read fraction*) instead receive per-item LLM reads. Both arms use the same model.

3.6 Two-queue triage (replacing single-score anomaly ranking)

Section 6.3 shows that no single geometric score separates the two things an anomaly panel conflates. The replacement uses two Euclidean signals: *novelty* = distance to the *nearest* leaf

centroid over all leaves (far from every known concept ⇒ the taxonomy lacks this concept), and *misassignment margin* = the amount by which some other leaf's centroid beats the item's own (strongly attracted elsewhere ⇒ filed in the wrong node). Each feeds its own queue: new-concept proposals to the anchor loop; reassignment candidates to review.

4 Experimental setup

Corpora. LEDGAR [16] (60,000 contract clauses, 100 provision types; the governing-law extraction subset has 3,167 clauses); 20 Newsgroups (6,000 posts, 6 super-groups / 20 groups); Amazon products (3,000 items, 6 top categories); WordNet nouns (82,115 synsets) as a known-answer control for hierarchy representation.

Models. Embedder: `nomic-embed-text` (768-d, local). Namer/extractor for the original experiments: `deepseek-v4-flash` at temperature 0. The audit suite (Sections 5.2–5.9) was run with a local `gemma3:27b` as the extractor in *both* arms of every paired comparison; its read-everything accuracy (97.0%) and real cluster-level accuracy (92.2%) match the DeepSeek figures (96.5%, 92.2%), so the pipeline is not extractor-fragile, and paired comparisons remain internally valid.

Ground truth. For governing-law extraction, a regex over the clause text (the state is literally written). Section 5.5 audits this oracle. Reference labels for purity/NMI/AMI come with each corpus and are evaluation-only.

Measurement rules (held throughout): significance for any accuracy comparison via paired bootstrap on per-item differences (4,000 resamples); if the CI includes zero the claim is "matches," never "beats"; purity (per-item %), NMI, and AMI reported as distinct quantities; chance-corrected AMI alongside NMI; majority and random-at-equal-budget baselines; never assume an LLM read is correct—measure it; a failed positive control invalidates the apparatus, not the theory. Random seeds fixed; seed sensitivity measured explicitly (Section 5.9).

5 Results

5.1 Cohesion is a label-free confidence signal

Across 83 LEDGAR leaves, cohesion correlates with gold purity at Spearman +0.74 (multivariate OLS over six geometric signals: in-sample $R^2 = 0.61$). Operationally, above-median-cohesion clusters are 98.0% accurate under cluster-level extraction versus 79.2% below the median: the gate separates the errors without labels.

Generality and the size confound (audit). Cluster size correlates with purity at -0.55 in the same probe, and small clusters are mechanically tighter, so cohesion could be a size proxy. Rebuilding structure-only trees on three corpora with five seeds each:

corpus	full ρ	partial ρ (| log size)	R^2 in-sample	R^2 LOO
LEDGAR	+0.711	+0.671	0.533	0.496
20 Newsgroups	+0.516	+0.483	0.345	0.247
Amazon	+0.736	+0.709	0.588	0.294

The univariate signal generalizes and survives the size control. The multivariate model does not survive leave-one-out on the smaller corpora; we therefore recommend the univariate cohesion rank and treat the R^2 as bookkeeping.

5.2 Gated extraction: the honest frontier

LEDGAR governing-law → US state; 2,475 clauses with oracle states; K=40 clusters; paired sample n=400.

Under the original protocol the cluster arm used the cluster's gold-majority label (an oracle). The audit replaced it with real per-cluster LLM calls. The oracle turned out to be nearly immaterial to the cluster arm itself, because real calls reproduce the majority label almost exactly (92.2% vs. 92.2% at seed 13; max gap 1.0 point over five seeds). What the honest test changed was the confidence interval story:

read %	calls	hybrid acc.	95% CI vs. read-all	at margin −2.0
10%	80	95.8%	$[-3.8, +1.2]$	not established
20%	120	96.5%	$[-2.8, +1.8]$	not established
30%	161	96.8%	$[-2.5, +1.8]$	not established
40%	206	97.8%	$[-1.0, +2.8]$	**established**
50%	242	98.0%	$[-0.8, +2.8]$	**established**

Read-everything on the same sample: 97.0% at 400 calls. With every prediction real, non-inferiority at a pre-declared −2.0-point margin holds from a 40% read budget (∼2× fewer calls) but is *not established* at the 20% budget that the original "5× fewer calls" headline used: the point estimate is −0.5 but the interval is too wide at n=400. *These are LEDGAR governing-law numbers*; Section 5.4 shows they sit at the favorable end of a task-dependent range and must not be quoted as the method's ratio.

Gate ablation at equal budget (all gates zero-call): cohesion beats a random gate at every budget (e.g. 94.7% vs. 92.4% at 100 reads) and per-item margin overtakes cohesion at high budgets (99.4% vs. 98.6% at 1,000 reads). The deployable insight is the zero-call *cascade*, with the signal chosen by budget, rather than cohesion specifically. A FrugalGPT-style gate needs one call per item (2,475-call floor here) and thus cannot operate in this regime at all.

5.3 Out-of-sample calibration (audit)

Choosing the read fraction on the evaluation sample is post-hoc selection. Splitting the 400 items into calibration/evaluation halves and selecting, on calibration, the smallest fraction matching read-everything: the procedure picked 10%; on the untouched half the policy scored 94.0% vs. read-everything 96.5% (CI $[-6.5, +1.5]$): non-inferiority *fails* out of sample, though the gate still beats a random gate at equal budget by +3.2 points. Small calibration sets overfit the threshold; a read-floor (≥30%) or larger calibration set is required. We report this as a genuine procedural weakness that the original evaluation could not see.

5.4 The parity budget is a corpus property (pre-registered falsification)

The governing-law task is unusually compressible: its clusters are nearly pure, so one propagated answer covers many items correctly. To test whether the frontier claim (B2) generalizes, we ran the identical protocol—real cluster-level calls, real per-item reads, random-gate control, calibration/holdout split, same model in both arms—on two task–corpus pairs never used during development: 20 Newsgroups topic assignment (6,000 posts, 20 labels, K=40) and Amazon top-category assignment (2,500 products, 5 labels, K=24). Falsifiers were declared before running:

B1-gate dies if the cohesion gate fails to beat random at every budget; B2 dies (in unconditional form) if no sub-100% budget reaches parity at the −2.0 margin.

	LEDGAR gov-law	20 Newsgroups	Amazon categ
read-everything	97.0%	67.0%	65.2%
cluster-first (propagation ceiling)	92.2%	57.2%	44.5%
ceiling gap	4.8	9.8	20.7
best hybrid ≤ 50% reads	98.0%	67.5%	57.5%
parity budget (margin −2.0)	40%	none ≤ 50% (CI $[-2.8, +3.8]$ at 50%)	none ≤ 100
gate vs. random at 30% reads	+3.1	+2.7	+3.7

Outcome. B1-gate survives: the cohesion gate beats the random gate at essentially every budget on both new tasks (e.g. +4.0 points at 10% reads on 20NG). Unconditional B2 is *falsified*: 20 Newsgroups is borderline at a 50% budget (point estimate +0.5, CI lower bound −2.8 just missing the margin at n=400), and Amazon reaches parity nowhere below reading everything. The determinant is visible in the table: the attainable parity budget tracks the *ceiling gap*—the distance between cluster-level propagation accuracy and per-item accuracy. When clusters are nearly pure (LEDGAR), reads only need to patch a small deficit and large savings follow; when the clustering does not align with the target labels (Amazon), no amount of gated patching short of reading everything closes the gap. The determinant is alignment, not corpus size: Amazon (2,500 items) and the LEDGAR subset (2,475 items) are the same size and sit at opposite extremes, so the condition cannot be stated as a minimum-N rule. Size enters as a separate multiplier: at parity the call ratio is approximately $N/(K + fN)$, so alignment sets the required read fraction f while corpus size amortizes the K cluster calls and raises the ceiling N/K; large, well-aligned corpora therefore support ratios far above the figures reported here, and large misaligned corpora support none. The surviving form of B2 is therefore conditional: *gated routing reaches parity at materially fewer calls exactly when the propagation ceiling approaches per-item accuracy, and the ceiling gap—itself measurable from a handful of calls—predicts the attainable ratio.* Any "$k\times$ fewer calls" claim for this family of methods is meaningless without reporting the corpus's ceiling gap; our own earlier 5× framing made precisely this error.

5.5 Auditing the oracle

The regex oracle silently dropped 519 of 3,167 clauses (16%) that plainly name a state (case and format brittleness), and the evaluation kept only regex-resolvable items, risking survivor bias toward easy items. A case-insensitive oracle (100.0% agreement with the original on their overlap) recovers 451 clauses. The audit's surprise: the recovered items are *easier*, not harder—cluster-first accuracy on them is 97.1% vs. 92.2% on the original set (pooled 92.9%), and per-item reads score 96.0%. The exclusion did not inflate the claim. 285 clauses (9%) remain oracle-less and would require human adjudication.

5.6 The end-to-end audit: finding is not free

Every number above conditions on the governing-law clauses being pre-isolated—by the *benchmark's gold labels*. In production the system must find the concept itself. Running production-shaped (take the tool's own category-blind tree, select the nodes its namer called "Governing Law," extract on that found set):

routing recall (true GL clauses found)	0.882
routing precision (found set purity)	0.752
extraction accuracy on correctly-found items	97.9%
end-to-end correct-value coverage	**86.4%**
false-assertion rate on intruder items	**88.9%**

Two distinct harms. First, coverage: 12% of true items are never seen, so the isolated-set 96–97% becomes 86.4% of contracts correctly filled. Second, and worse, *record poisoning*: cluster-level propagation hands the cluster's state to the 25% of found items that are not governing-law clauses at all (venue and jurisdiction lookalikes), writing ∼820 confident wrong values at corpus scale. The benchmark structurally cannot observe either harm. The mitigation is deterministic and free: validate every propagated value against the item's text (does the clause contain the asserted state or governing language?); failures route to *unresolved*, never to a silent value. Until such a validator runs, isolated-set accuracy must not be quoted as system accuracy.

5.7 Human effort and cross-domain behavior

On 20 Newsgroups, naming 20 clusters organizes 6,000 documents at 57.3% per-item purity (NMI 0.572, AMI 0.567), roughly 300× fewer human decisions than per-item labeling; purity rises and NMI/AMI peak exactly at the true group count, giving a label-free depth heuristic. The same pipeline behaves consistently on legal (purity 8–10× majority baseline), news (above), and product text (68.9% vs. 20% baseline). EMERGE cuts the false-merge rate from 66.2% to 15.9% at a purity cost (41.6 → 29.9); we flag honestly that the false-merge metric shares its seam signal with the split rule (the router grades its own homework), so the magnitude awaits blind human adjudication.

5.8 Naming

A minimal prompt ("shortest canonical name") beat a verbose one on every measure: mean words 3.06 → 1.40, names containing "and" 35% → 2%, similarity to the canonical legal heading 0.766 → 0.886, duplicate names per repeated concept 2.29 → 1.35. A compound-name resolution heuristic (pick the conjunct closest to the centroid) *lost* (0.682 vs. 0.733) and was not shipped.

5.9 Seed stability

Across five clustering seeds with real cluster calls per seed, hybrid accuracy at fixed budgets spreads 2.5–2.8 points, driven by cluster-arm variance (real cluster-first 90.8–93.5%). Gate numbers on this task should be read as ±1.4 points.

6 Negative results and corrections

We report these at equal weight; several killed our own hypotheses.

6.1 Pipeline negatives

Crystallization feedback (re-embed the discovered tree, reassign, iterate) degraded monotonically (leaf NMI 0.522 → 0.172 over three rounds); note this loop used the gradient apparatus of

Section 6.2, so its honest status is apparatus-suspect rather than falsified. *Whitening* destroys category structure (68.9% → 45.4%): the top principal components *are* the signal, while plain mean-centering fixes embedding anisotropy for free (mean category-pair cosine 0.91 → −0.24) with no clustering change [15]. *Cluster-first extraction fails on incidental attributes*: an attribute mentioned in passing (payment periodicity) scored 33.9% against an 80.1% majority baseline; the method requires the attribute to dominate the text. *Annealed refinement* hurt (−2.7 purity) where greedy cohesion refinement helped in MAP mode (+5.2); refinement also raises the false-merge rate, so it should be mode-coupled.

6.2 The hyperbolic arc: the optimizer was the problem

We initially published (internally) that hyperbolic embeddings store a 600k-node hierarchy without collapse. **Retraction 1:** evaluation pairs were sampled uniformly; in a broad tree nearly all random pairs meet at the root (643 of 800 pairs shared one identical tree distance), so the correlation target was near-constant and the measurement meaningless. Stratified sampling plus a validity gate ($\geq$4 distinct target values, std $\geq$0.5) flipped the result to "Euclidean wins everywhere." **Retraction 2:** that conclusion was withdrawn as well when the hyperbolic arm failed a positive control: on WordNet, where hyperbolic embeddings famously excel [12, 13], our gradient-trained arm scored parent-reconstruction MRR 0.391 (literature ≈0.98), with 88.6% of points pinned at the ball's projection ceiling and radius–depth correlation +0.025, *while training loss fell monotonically.* The apparatus, not the geometry, was broken; the six clustering losses we had attributed to hyperbolic geometry are equally void. Two of the retracted storage rows carry a second, independent defect: the "real discovered" hierarchies were KMeans-imposed on product embeddings with no verified intrinsic hierarchy, so they tested storage of an artificial tree.

Resolution: replacing optimization with Sarkar's deterministic construction [14, 13] (children placed by Möbius isometries at hyperbolic step τ; no loss function) passes the control decisively: on 82,115 WordNet synsets in *two* dimensions, radius–depth +0.774 and tree-distance $\rho = +0.864$, beating our gradient-trained Euclidean at eight dimensions (+0.610). A τ sweep reproduces the precision wall Sala et al. document (τ=3.0 → 0.880; τ=5.0 → 0.626 as float64 exhausts near the boundary).

The transferable rule: *gradient-trained Poincaré embeddings must pass a known-answer control before any comparative claim, and falling loss is not evidence of health.* Under this rule, hyperbolic geometry in our system retains exactly one validated role—compact representation and visualization of an already-built tree—and no valid losses.

6.3 Falsifying our own anomaly score

The shipped novelty detector ranked items by Poincaré misfit to their leaf centroid plus 0.15× ball radius, on a PCA-projected ball. A leaf-level correlation had suggested redundancy with cohesion; that is the wrong test for an item-level ranker, so we re-tested at item level with the innovation steelmanned (the radius term as a distinctiveness weight) and vindication conditions declared in advance. Ground truth from gold labels distinguishes *novel* items (concept dominant in no leaf; 19,217 of 60,000) from *misassigned* items (concept lives elsewhere; 16,765).

Result: plain cosine misfit beats the shipped score on every metric (anomaly AUC 0.814 vs. 0.763; novel precision@40 17.5% vs. 5.0%), and the rankings genuinely differ (top-40 overlap 11/40). The radius term contributes ∼3% of the score's spread, and radius alone is *anti-*

correlated with anomaly (AUC 0.335). Inspecting per-type mean radii explains why: the ball's center-to-edge axis reads *abstract-to-concrete* (central: Authorizations, Binding Effects, Authority; peripheral: Base Salary, Vacations, Counterparts), and the taxonomy's homeless concepts are disproportionately the abstract ones—so the shipped +radius weight pointed *away* from novelty. Inverting the sign lifts the misfit score's novel yield to 37.5%@40, but the best signal needs no radius at all: distance to the *nearest* leaf reaches 55% novel@40 (11× the shipped score) and margin-to-another-leaf reaches 100% misassigned@40, motivating the two-queue design of Section 3. Scope: this falsifies the shipped PCA-ball implementation; no item-level hyperbolic embedding that passes a depth control currently exists to test.

7 Limitations and threats to validity

(1) The false-merge metric shares its signal with the EMERGE split rule; blind human adjudication is the open control. (2) One embedder; cohesion thresholds are not calibrated across embedding models. (3) The audit-suite extractor differs from the original DeepSeek runs (with strong convergence, but exact-model replication is pending), and on Amazon the cluster-labeling step must map product text onto catalog category names, a mismatch that depresses the propagation ceiling; the falsification stands because both arms share the label vocabulary, but the ceiling gap there partly measures label-vocabulary fit. (4) n=400 paired samples bound the resolvable margin; the 20 Newsgroups 50%-budget case is decidable with $n\approx1000$. (5) The regex oracle, though audited, still cannot score 9% of items. (6) Contract-level rollup from clause-level results is untested. (7) Seed variance of ±1.4 points bounds the precision of all gate numbers. (8) LLM outputs are provider-version dependent even at temperature zero.

8 Conclusion

A zero-call geometric signal can decide where language-model inference is worth spending: cohesion's predictive value replicates across corpora, survives a size confound, and its gate beats random routing on tasks never used to tune it. What does *not* generalize is any particular savings number: our own 5× figure was the property of one unusually compressible corpus, and the pre-registered falsification test showed the attainable parity budget is governed by the ceiling gap between cluster-level propagation and per-item accuracy—a quantity measurable for a few dollars before committing to the method. The audits matter more than the headlines: an oracle everyone should distrust turned out immaterial; a threshold procedure everyone would have shipped failed out of sample; a benchmark that looked complete was silently subsidized by its own gold labels, hiding both a coverage loss and a record-poisoning failure mode with a free deterministic fix; and a savings constant that looked like a property of the method was a property of the corpus. The hyperbolic arc contributes a reusable methodological rule (control first; loss is not health) and one validated capability (constructed 2-D representation of finished hierarchies). We offer the pattern—claims stated at the right altitude, pre-registered falsifiers, paired tests, positive controls, and adversarial audits of one's own headline—as the paper's most transferable result.

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